

Rethinking Sleep Staging

Unsupervised cross-laboratory mouse sleep staging with temporal variational autoencoders
and substage discovery

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Abstract

Cross-laboratory manual scoring of mouse sleep from EEG and EMG lacks consensus, and supervised models trained at one site often fail on others without retraining or per-animal calibration. On a quality-screened MSSV cohort (20 wild-type mice, laboratories 2/3/5, four-fold leave-mice-out cross-validation), decoder-only conditional variational autoencoders stage held-out animals from spectral inputs alone, without subject identifiers or manual baseline scoring at inference. A sticky HMM-GMM prior reaches mean prior normalised mutual information (NMI) 0.575 on joint cross-laboratory holdout versus 0.282 for a matched static mixture baseline (paired mean $\Delta = +0.29$ across four folds; one-sided Wilcoxon $p = 0.063$). After establishing cross-laboratory holdout staging, a resolution sweep asks whether Wake, NREM, and REM alone describe the data: at $K=4$ on held-out fold 4, substates carry distinct spectra, bout structure, and latent geometry beyond a single cluster per macro class, in line with finer unsupervised structure reported in rodents [1, 2]. The framework supports cross-study comparison without harmonised expert labels but does not replace supervised scoring or recover REM reliably on strict holdout (REM recall $\sim 3\%$).

Introduction

Rodent sleep is scored from EEG and EMG into Wake, non-rapid eye movement (NREM), and rapid eye movement (REM). That three-class vocabulary is the standard benchmark, but it compresses continuous physiology into coarse bins. Unsupervised work in mice has repeatedly resolved substages with distinct spectra, bout lengths, and transition roles that Wake/NREM/REM labels alone do not name [1–3]. Montage and laboratory protocols nonetheless differ across studies [4], so any claim that sleep structure extends beyond macro scoring must survive strict holdout on new animals and new sites. Rose et al. report limited cross-laboratory scoring consensus and poor transfer of published deep models across MSSV sites unless retrained on multi-laboratory data [5]. Supervised pipelines such as SPINDLE and REST can reach high accuracy under that regime but depend on expert labels that inherit rater noise [6, 7]. Transductive tools including AccuSleep require roughly ten minutes of manually scored baseline per new mouse [8, 9]. These constraints motivate label-agnostic staging under strict population holdout without per-animal calibration at inference.

Unsupervised sequence models infer latent structure directly from EEG and EMG dynamics and align naturally with bout persistence and transitions [3]. Static clusterers and shallow HMMs discover states from raw or hand-crafted features but rarely evaluate zero-shot transfer to held-out mice from other laboratories under a shared population model [10–12]. Katsageorgiou et al. resolved rodent substages beyond the macro taxonomy with distinct physiology and bout structure (mcRBM) [1], yet without cross-laboratory leave-mice-out holdout. García Ciudad et al. showed that longer temporal context improves supervised rodent staging, especially for REM [13]. That setting remains fully supervised and does not test label-free cross-site transfer. In a single-laboratory cohort with encoder+decoder conditioning and expert features, mixture-prior cGMVAE reached population prior NMI of about 0.70 under leave-one-subject-out prediction [2]. Cross-laboratory holdout on spectral CNN inputs is substantially harder and is the focus here.

We extend that line with decoder-only conditional variational autoencoders and sticky HMM-GMM mixture priors. Subject embeddings condition reconstruction during training only, encoder latents remain free of identity information, and held-out mice are staged from the learned prior without subject identifiers [14–17]. The report first validates decoder-only, cross-laboratory holdout staging at $K=3$ as a methodological prerequisite. It then tests whether the latent data support finer structure than Wake, NREM, and REM through mixture-resolution sweeps and physiology-first interpretation [2]. Contributions are threefold. (i) Decoder-only conditioning with zero-shot holdout staging. (ii) A fair cross-laboratory ladder comparing static and temporal mixture priors on locked spectral inputs. (iii) Evidence that unsupervised substates, with separated spectra, latent geometry, and transition structure, persist under holdout and motivate rethinking staging beyond the three-class grid. We analyse 20 quality-screened mice from MSSV laboratories 2, 3, and 5 (Bern, Copenhagen, Lyon) under four-fold leave-mice-out cross-validation, comparing a static Gaussian-mixture prior with a warm sticky HMM-GMM prior on matched inputs, then sweeping mixture resolution on held-out fold 4 and, complementarily, across the full cohort for substage physiology panels in the thesis tradition (Figs 27–30 therein [2]). Dataset and protocol details are in Data and Appendix A. Training detail, prior equations, and ablations are in Appendices B–C.

Data

Dataset and cohort. MSSV (OpenNeuro ds006366) [4] comprises invasive EEG and neck EMG from 92 mice across five European laboratories, with expert Wake, NREM, REM, and Artifact labels on a 4s grid for evaluation only. We analyse laboratories 2 (Bern), 3 (Copenhagen), and 5 (Lyon): healthy wild-type sites with complementary montages, excluding laboratories 1 and 4 (pathology-specific or single-electrode cohorts). Quality screening retains mice with low EEG drift (relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode) and EMG that separates Wake from NREM (Wake/NREM root-mean-square ratio ≥ 2 ; full criteria in Preprocessing and spectral inputs), yielding a 20-mouse cohort (6, 10, and 4 per site, ~ 453 h, four-fold leave-mice-out). The subset balances cross-laboratory coverage with stable spectral inputs. Training mice never appear in the validation fold for a given split (S9 Table). No new animal procedures were performed.

Preprocessing and inputs. Montages are harmonised to a parietal-frontal EEG differential plus neck EMG (lab 2: EEG1 and EEG4, labs 3/5: EEG1 and EEG2). Per-site spectral filters and band limits were fixed from incohort ablations (Preprocessing and spectral inputs, Spectral bands and frequency content). Recordings are segmented into 4s epochs at 128 Hz. Artifact epochs are removed. Each epoch is a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$, batched into contiguous sequences for temporal models without hand-crafted expert features.

Methods

Conditional variational models and latent priors

Spectral EEG/EMG epochs are mapped to a low-dimensional latent space by a convolutional variational autoencoder (cVAE). A subject embedding conditions reconstruction during training only. At holdout, staging uses encoder means and a learned mixture prior without subject identifiers. We compare two prior structures on matched inputs and cross-validation folds: a static Gaussian mixture (cGMVAE, $T=1$) and a warm sticky HMM-GMM sequence prior (cHMM-GMVAE, $T=64$ epochs, ~ 4.3 min), with $K=3$ for macro staging. HMMGMVAE without subject conditioning provides a sequence-prior baseline (Table 1) [17]. Locked recipes, ablation rationale, and sensitivity checks are in Training, checkpoints, and metrics.

Encoder-decoder. Training batches are contiguous windows $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times C \times F}$ with $T = 64$. A convolutional encoder maps each epoch to a diagonal Gaussian posterior,

$$q_\phi(\mathbf{z}_t \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{z}_t \mid \boldsymbol{\mu}_t, \text{diag}(\boldsymbol{\sigma}_t^2)), \quad (\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2) = f_\phi(\mathbf{x}_t), \quad (1)$$

with f_ϕ a CNN-MLP encoder and $\mathbf{z}_t \in \mathbb{R}^d$, $d=6$. The decoder reconstructs each epoch from $\mathbf{z}_t$ and, during

training only, a subject embedding $\mathbf{e}_i \in \mathbb{R}^8$:

$$\hat{\mathbf{x}}_t = g_{\theta}(\mathbf{z}_t, \mathbf{e}_i). \quad (2)$$

Under decoder-only conditioning, f_{ϕ} does not receive $\mathbf{e}_i$ (Decoder-only vs encoder+decoder conditioning). Architecture widths are in Training, checkpoints, and metrics and Fig 1.

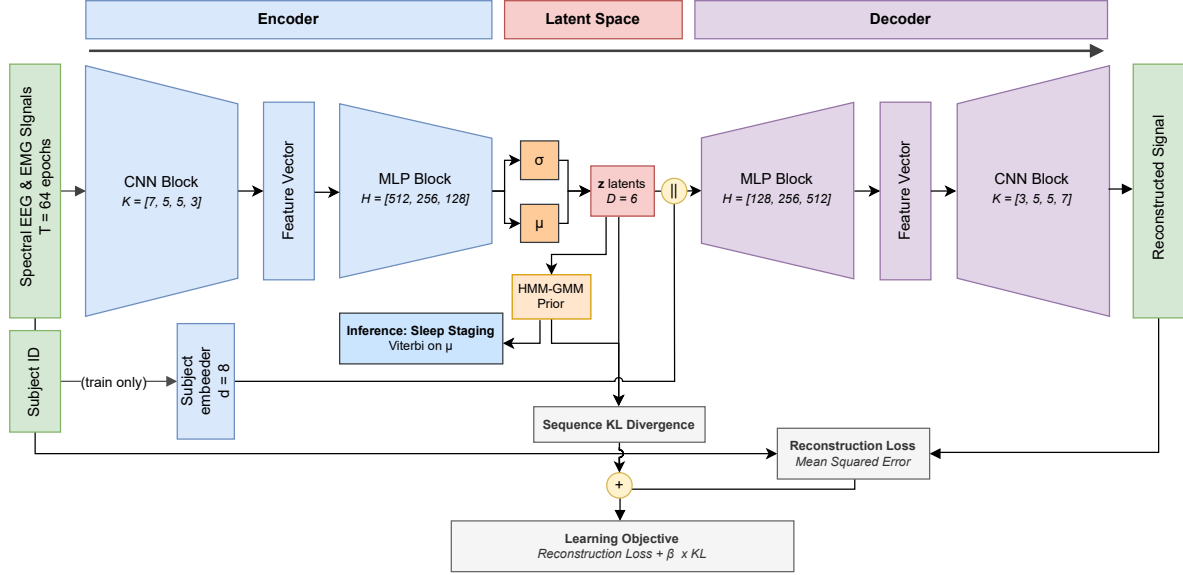


Fig 1. Conditional GMVAE architecture. **Encoder:** spectral EEG/EMG maps pass through CNN ($K=[7, 5, 5, 3]$) and MLP blocks. Subject ID feeds a subject embedder used only in the decoder. **Latent:** diagonal Gaussian posterior with dimension $d=6$ feeds a Gaussian-mixture (cGMVAE) or sticky HMM-GMM (cHMM-GMVAE) prior. **Decoder:** latent vector concatenated with subject embedding, then CNN reconstruction. Training minimises MSE reconstruction plus $\beta \times$ prior KL. At holdout, only the encoder and prior are used for staging (no subject ID, no decoder).

Variational objective and prior. Training minimises reconstruction error plus a β -weighted KL term ($\beta=0$ for ten epochs, then linear ramp 0.01–1.0; schedule in Training, checkpoints, and metrics). cGMVAE uses a static Gaussian mixture prior. cHMM-GMVAE uses a sticky HMM-GMM prior ($\kappa=0.92$) with phased warm-start from encoder means (HMM-GMM prior (equations), Training, checkpoints, and metrics) [16, 17].

Prior-based staging at holdout. Held-out mice are staged from encoder means μ_t and the trained prior without subject embeddings or decoder reconstruction (Training, checkpoints, and metrics). Static cGMVAE assigns $\hat{s}_t = \arg \max_k [\log \pi_k + \log b_k(\mu_t)]$. cHMM-GMVAE applies Viterbi decoding,

$$\hat{\mathbf{s}}_{1:T} = \arg \max_{\mathbf{s}_{1:T}} \left[\log \pi_{s_1} + \sum_{t=2}^T \log A_{s_{t-1}, s_t} + \sum_{t=1}^T \log b_{s_t}(\mu_t) \right], \quad (3)$$

with $B_{tk} = \log b_k(\mu_t)$.

Cross-validation and holdout protocol

Evaluation scopes. The 20 quality-screened mice were partitioned into four folds with leave-mice-out cross-validation (S8 Table, S9 Table). Each fold holds out one to three mice per site (four to six mice total). The remaining 14–16 mice form the training pool and never overlap the validation set. Joint scope pools all training-fold mice from laboratories 2, 3, and 5 into one population model and tests on the held-out fold

across sites. Within-lab scope restricts both training and validation to a single laboratory while using the same fold splits. Absolute prior NMI values between joint and within-lab scopes are not compared because the training distributions differ.

Holdout inference. Held-out mice are staged without subject identifiers and without decoder reconstruction. The encoder produces deterministic means μ_t from spectral inputs only. Subject embeddings and the decoder are not used. State labels are assigned from the trained GMM or sticky HMM-GMM prior on μ_t (Eq. (3), prior details in HMM-GMM prior (equations)). Per-laboratory spectral locks remain fixed at holdout.

Agreement metrics. Agreement with expert Wake, NREM, and REM labels was quantified primarily using normalised mutual information (NMI),

$$\text{NMI}(Y, \hat{Y}) = \frac{2 I(Y; \hat{Y})}{H(Y) + H(\hat{Y})}, \quad (4)$$

where I is mutual information and H Shannon entropy. Macro-aligned accuracy, Cohen’s κ , and per-class recall are reported in S4 Table and S10 Table. Definitions are in Training, checkpoints, and metrics.

Checkpoints and inferential testing. Training checkpoints are selected without expert labels using a composite validation score [2] (Eq. (5), Training, checkpoints, and metrics). Each fold trains three seeds. We report the best holdout prior NMI across seeds (Tables 1–2) and paired cHMM-GMVAE vs. cGMVAE tests (S12 Table, S8 Fig).

Training, substages, and reproducibility

Decoder-only conditioning. Subject identity enters only the decoder path (Eq. (2)), not the encoder (Eq. (1)). Incohort ablations favoured decoder-only over encoder+decoder conditioning on all three laboratories (Decoder-only vs encoder+decoder conditioning). Nuisance amplitude and montage effects are absorbed during reconstruction while encoder means remain free of identity information for holdout staging—the design choice that separates this report from the thesis encoder+decoder recipe [2].

Training and checkpoints. Models are trained on all mice in the training folds. The best unsupervised checkpoint per seed is retained (Eq. (5)). Macro ladder comparisons use $K=3$. A substage sweep varies $K \in \{3, 4, \dots, 15\}$ on joint holdout fold 4. We selected $K=4$ as the highest K within one standard deviation of the peak mean NMI with non-degenerate occupancy (Fig 2). Physiology summaries follow the thesis substage workflow [1, 2].

Data and code availability. MSSV is available on OpenNeuro (ds006366). Processed features, analysis code, and full experimental settings are provided as supplementary material (Supplementary experimental settings).

Results and discussion

Cross-lab holdout performance

Under joint population training with leave-mice-out four-fold cross-validation across MSSV laboratories 2, 3, and 5, decoder-only conditional models were evaluated on held-out mice staged without subject identifiers (Table 1). Supervised ceilings calibrate expectations (S1 Table). SPINDLE reaches high cross-laboratory accuracy under supervised retraining [6]. REST reports $\kappa \approx 0.87$ on multi-strain mouse cohorts [7]. García Ciudad et al. show that extending supervised context to ~ 28 s improves staging over single-epoch classifiers, with the largest REM gains [13]. Our best joint holdout prior NMI (0.575 mean for cHMM-GMVAE) is far below those supervised κ /accuracy values, as expected for zero-shot, label-free transfer without per-mouse calibration, but the relative lift over matched cGMVAE (+0.29 NMI) isolates the value of temporal priors under the same unsupervised constraint.

In Table 1, cHMM-GMVAE exceeds cGMVAE on every fold (mean prior NMI 0.575 vs. 0.282). Paired fold-level analysis yields mean $\Delta\text{NMI} = +0.29$ (bootstrap 95% CI [0.19, 0.36]; one-sided Wilcoxon $p = 0.063$; paired t -test $p = 0.011$; S12 Table, S8 Fig). The gain is positive on all four folds and persists when using seed means rather than best-of-three (mean $\Delta = +0.26$, Wilcoxon $p = 0.063$). Both models extract structure above

Table 1. Joint holdout prior NMI by fold (best-of-three seeds, locked recipes, $K=3$). HMM (raw) folds 1–3: best logged validation prior NMI from incomplete joint runs (≤ 0.003); fold 4 from full holdout.

Fold	HMM (raw)	cGMVAE	HMMGMVAE	cHMM–GMVAE
1	0.003	0.267	0.409	0.579
2	0.001	0.256	0.236	0.396
3	0.003	0.307	0.497	0.675
4	0.208	0.299	0.404	0.649
Mean	0.054	0.282	0.387	0.575

chance-level macro agreement, but the temporal prior adds a consistent margin. This matches supervised evidence that mouse staging benefits from context beyond the current epoch, especially for REM [13], here under label-free training and strict mouse holdout. The margin was largest on folds 3 and 4 (+0.36 and +0.41 NMI) and smallest on fold 2 (+0.10), where difficult Bern montages mix with held-out mice from all sites. Per-mouse fold-4 analysis (S2 Fig) shows the improvement is distributed across animals (paired Wilcoxon on six held-out mice: $p = 0.016$, mean $\Delta = +0.35$, S12 Table), not driven by a single outlier. Macro confusion on fold 4 (S3 Fig, S10 Table) confirms strong Wake and NREM alignment ($\kappa \approx 0.81$) at $K=3$, but REM recall remains low ($\sim 3\%$), indicating residual rare-class difficulty under label-free training despite temporal context. Per-fold breakdowns including HMMGMVAE appear in S2 Table. Seed variance is in S11 Table. These $K=3$ results establish that label-free models can align with expert Wake/NREM/REM on held-out mice across laboratories. They do not imply that three classes are sufficient descriptions of the signal.

Laboratory heterogeneity

Table 2 compares holdout performance under joint population training versus within-laboratory training on the *same* held-out mice from each site. For joint training, NMI and macro-aligned accuracy are computed on laboratory-specific holdout subsets only (not pooled across sites).

Table 2. Fair per-laboratory holdout comparison (mean over four folds, $K=3$). Joint: train on all laboratories, report metrics on held-out mice from the listed site only. Within: train and evaluate within the same site. All rows use the same locked preprocessing per laboratory (spectral bandpass, montage, post-normalisation from incohort ablation winners, Preprocessing and spectral inputs) and matched CNN–VAE capacity ($d=6$, MLP widths 512–256–128, 200 epochs, batch 128, three seeds). cGMVAE, HMMGMVAE, and cHMM–GMVAE share learning rate 1.3×10^{-3} . They differ only in embedding use (8 vs. 0), sequence length ($T=1$ vs. $T=64$), and prior type (static GMM vs. HMM–GMM vs. warm sticky HMM–GMM). HMM (raw) uses the time-domain pipeline (no shared deep front-end). VAE models report best-of-three prior NMI and matching macro accuracy, averaged across folds. Per-fold within-lab and joint breakdowns: S2 Table. Incohort ceiling NMI from per-lab ablation sweeps (upper reference, not holdout): Incohort performance ceiling. HMM (raw): $runs=1$; joint per-site NMI for folds 1–3 unavailable (pooled joint NMI ≤ 0.003).

Lab	Model	Joint NMI	Joint Acc.	Within NMI	Within Acc.
Lab 2	HMM (raw)	0.179	0.665	0.004	0.500
	cGMVAE	0.223	0.648	0.549	0.855
	HMMGMVAE	0.039	0.543	0.586	0.865
	cHMM–GMVAE	0.570	0.869	0.596	0.864
Lab 3	HMM (raw)	0.237	0.731	0.029	0.565
	cGMVAE	0.415	0.796	0.516	0.844
	HMMGMVAE	0.586	0.850	0.637	0.897
	cHMM–GMVAE	0.705	0.923	0.738	0.939
Lab 5	HMM (raw)	0.240	0.724	0.001	0.498
	cGMVAE	0.280	0.715	0.476	0.853
	HMMGMVAE	0.483	0.821	0.584	0.882
	cHMM–GMVAE	0.556	0.877	0.567	0.885

Table 2 shows that within-lab training consistently exceeds joint training on the same holdout mice (largest gap on laboratory 2), while cHMM-GMVAE leads on laboratory 3 under both scopes. This pattern is consistent with residual covariate shift when pooling heterogeneous spectral profiles across sites. Joint holdout is weakest on laboratories 2 and 5 relative to Copenhagen: Bern contributes six mice (~ 73 h) with a four-electrode implant and wider EEG bandpass (0–30 Hz), Lyon four mice (~ 144 h, one very long recording), whereas Copenhagen contributes ten mice (~ 236 h) on the shared two-EEG montage (S8 Table). Smaller cohorts, shorter per-mouse coverage, and site-specific hardware therefore leave less stable population structure for zero-shot transfer than at laboratory 3. Decoder-only subject embeddings index mice, not laboratories. They absorb nuisance amplitude during reconstruction but do not fully disentangle equipment and scoring context from identity, which partly explains why explicit lab-conditioning ablations did not improve joint holdout (Decoder-only vs encoder+decoder conditioning). On laboratory 5, HMMGMVAE without subject embeddings matches or exceeds cHMM-GMVAE within-lab, suggesting montage-specific gains from temporal priors rather than subject conditioning alone. Under joint training on laboratory 2, HMMGMVAE prior NMI collapses (0.039 mean), consistent with a sequence prior without subject embeddings failing on the Bern montage when pooled across sites. Holdout NMI in Table 2 remains below incohort ceiling NMI in Incohort performance ceiling because leave-mice-out validation is strictly harder than pooling all laboratory mice during tuning. HMM (raw) on time-domain EEG/EMG remains near chance under joint training (mean prior NMI 0.054 across four folds; folds 1–3 ≤ 0.003 , fold 4 0.21), underscoring the representation mismatch between raw voltages and stage-discriminative spectral structure on this protocol (cf. thesis Ch. 5).

Beyond Wake, NREM, and REM: substage resolution

Expert Wake, NREM, and REM are treated here as a *reference* partition for NMI, not as the hypothesis that sleep contains only three dynamical regimes. The scientific question is whether unsupervised models trained under the same cross-laboratory holdout constraints recover finer substages with interpretable physiology, as in mcRBM and in single-laboratory cGMVAE work [1, 2], rather than collapsing to three macro-like blobs.

On joint holdout fold 4 we swept $K \in \{3, 4, \dots, 15\}$ (Fig 2). At the ladder default $K=6$, holdout prior log-likelihood for the three model families is tabulated in S14 Table (with the sequence-length caveat noted there). Mean validation NMI is similar between $K=3$ and $K=4$ (0.581 vs. 0.563 across five and four seeds, respectively) and falls beyond $K=6$ (mean 0.516), so agreement with expert *macros* does not keep rising when more substates are allowed. That mismatch is itself evidence that NMI against Wake/NREM/REM under-rewards genuine finer structure. At $K=4$, two states dominate occupancy ($\sim 52\%$ Wake-dominant, $\sim 48\%$ NREM-dominant) while two rare REM-dominated clusters remain ($\sim 0.3\%$ and $\sim 0.2\%$, S3 Table). This is a modest split of the macro grid under strict holdout, not yet the rich transition taxonomy seen at higher K in less stringent settings. The dominant off-diagonal transition at $K=4$ is NREM $\rightarrow$ Wake ($p=0.95$, S5 Fig), consistent with brief arousals at NREM boundaries. At $K=13$ on the same fold, transition structure diversifies (0.525 NMI, S6 Fig), paralleling the thesis finding that higher mixture counts expose boundary-like and transition-dominant substates even when macro NMI plateaus [2]. Complementarily, a population-level sweep trains on all 20 cohort mice (incohort, not leave-mice-out) to support substage-resolved physiology and transition panels analogous to thesis Figs 27–30 [2]. Holdout fold 4 remains the generalisation benchmark. K for the holdout panels below was chosen on fold 4 only (Limitations).

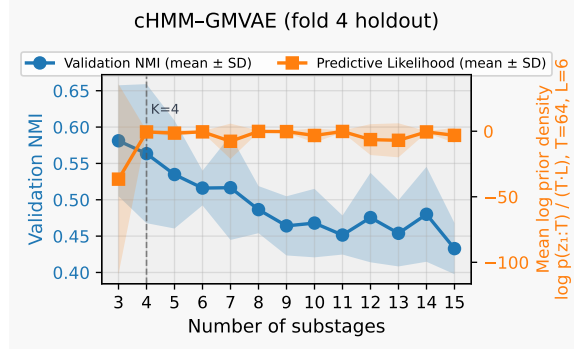


Fig 2. Substage resolution sweep (joint holdout fold 4). Mean validation prior NMI (blue) and mean log prior density $\log p(\mathbf{z}_{1:T}) / (T \cdot L)$ with $T=64$ latent time steps and $L=6$ latent dimensions (orange) versus mixture count K . Shaded bands: mean $\pm$ SD across seeds. Dashed line: chosen $K=4$.

Dynamics and transitions

Interpretation follows the thesis substage workflow [2]: (i) substage-resolved physiology (thesis Fig 27), (ii) transition structure between substages (Fig 28), (iii) temporal sequencing on hypnograms (Fig 29), and (iv) grouping into stable and boundary-like roles (Fig 30). Here we apply the same logic on cross-laboratory holdout at $K=4$.

Fig 3 pairs a transition taxonomy (**A**) with the substage physiology grid (**B**; fold 4 holdout, best seed). Panel **A** places substages at latent PCA centroids with thresholded transition edges [3]. Panel **B** summarises per-substage spectra, θ/δ , bout length, expert macro composition, and subject mix. These are the quantities a sleep biologist uses to judge whether a discovered state is meaningful beyond its macro label. Wake- and NREM-dominant substages show separated spectral profiles. Rare REM-dominated components carry $< 1\%$ occupancy yet occupy distinct regions of latent space (Fig 4), supporting the view that expert REM epochs are heterogeneous and that a single REM bin merges physiologies that unsupervised models separate [1, 2]. Encoder means at $K=4$ form separated clusters rather than a single mixed cloud, indicating that the VAE prior discovers structure not reducible to plotting Wake, NREM, and REM alone. Hypnogram exemplars (S5 Fig) show bout-consistent predicted trajectories on held-out mice. Population-level sweeps at higher K are intended to extend these panels with fuller transition taxonomies under biological review.

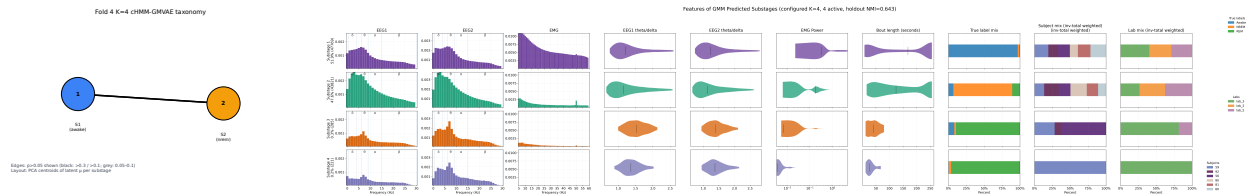


Fig 3. Substage organisation and physiology (fold 4 holdout, $K=4$). **A**, Transition taxonomy: latent PCA layout and significant edges between substages. **B**, Physiology grid (best seed): EEG/EMG spectra, features, true macro mix, and subject mix per discovered substage.

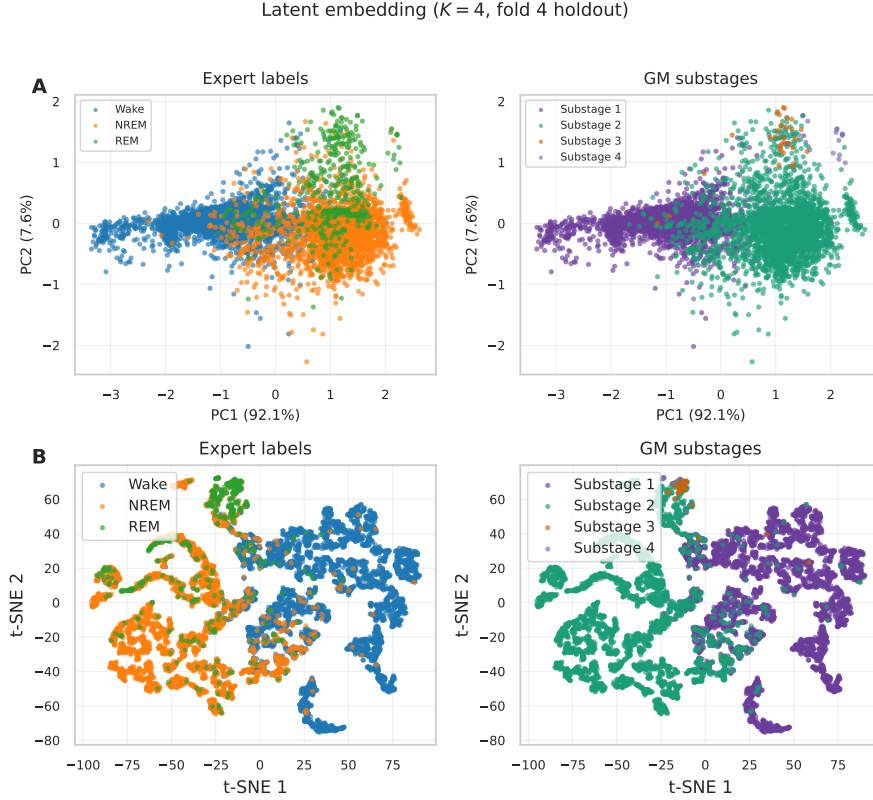


Fig 4. Latent embedding ($K=4$, fold 4 holdout, best seed). PCA of encoder means μ_t , coloured by predicted substate on held-out mice.

Methodological implications

The intellectual arc differs from staging papers that optimise accuracy against Wake/NREM/REM alone. Cross-laboratory holdout first establishes that the representation is usable without labels or calibration. The substage panels then ask what structure those latents encode when mixture resolution increases [1, 2]. Low or flat macro NMI at $K > 4$ should not be read as failure if substages remain physiologically separated. Expert macros are a noisy, coarse map of a finer landscape [5]. The evaluation ladder mirrors the thesis progression from controlled validation to population transfer, but shifts the primary metric to cross-laboratory leave-mice-out holdout on spectral CNN inputs rather than in-laboratory leave-one-subject-out on expert features [2]. Decoder-only conditioning was favoured over encoder+decoder paths in incohort ablations on all three sites (Decoder-only vs encoder+decoder conditioning), trading in-laboratory NMI for latents that remain usable without subject identifiers at inference. Temporal mixture priors help where montages and cohort size support sequence modelling (folds 3–4, Copenhagen within-lab). They do not uniformly dominate. Static or unconditioned sequence baselines can match cHMM-GMVAE when subject embeddings are unavailable and montage heterogeneity is high (laboratory 5 within-lab, laboratory 2 joint HMMGMVAE collapse). Complementary MSSV work on cross-species alignment (QSLP-AE) targets human–mouse representation matching rather than zero-shot cross-laboratory mouse holdout [18]. The present framework is intended for label-free comparison of vigilance structure across acquisition sites without harmonised expert scoring.

Limitations

Agreement is quantified by NMI against expert macro labels, which inherit cross-laboratory disagreement [5]. High NMI therefore measures discovery of shared structure, not mimicry of a single rater. Appendix macro

accuracy can read higher than NMI implies because dominant-label mapping rewards macro-pure clusters even when cluster boundaries differ from expert scoring. REM recall on fold 4 remains near 3% despite strong Wake/NREM alignment. Temporal context alone does not close the REM gap under strict holdout. Cross-laboratory holdout remains harder than within-laboratory training on the same mice (Table 2), and seed or fold variance (fold 2, S11 Table) indicates remaining fragility. Inferential tests use only four paired folds and therefore have limited power (Wilcoxon $p = 0.063$ for the primary comparison despite $\Delta > 0$ on every fold). The substage sweep and $K=4$ selection use held-out fold 4 only. K may not generalise to other folds without train-only model selection. Reported NMI uses best-of-three seeds per fold, which is optimistic relative to seed means. Seed-mean sensitivity in S12 Table preserves the same direction. Fixed spectral front-end locks control laboratory nuisance but do not replace per-mouse calibration. Raw-signal CNN encoders and lab-aware conditioning are outlined in Future directions.

Conclusion

Wake, NREM, and REM remain indispensable for comparing studies, but they are an incomplete description of mouse sleep dynamics. Unsupervised models trained on MSSV recover substates with distinct physiology and latent organisation that the three-class grid does not explicitly represent [1, 2]. This report shows that claim can be pursued under strict cross-laboratory leave-mice-out holdout, not only within a single laboratory. Decoder-only cHMM-GMVAE improves zero-shot alignment with expert macros relative to matched cGMVAE (mean prior NMI 0.575 vs. 0.282; paired $\Delta = +0.29$ across four folds), establishing a label-free staging pathway across sites. Temporal priors and subject embeddings at training support that transfer without identity at inference. The substage-resolution sweep and physiology panels (Figs 2–4) carry the main scientific payoff. Mixture count should be chosen for interpretable structure and transitions, not only macro NMI. Holdout at $K=4$ yields a modest macro-plus split. Richer transition substages appear at higher K (S6 Fig) and in population-level analyses in the thesis tradition. The framework does not yet replace supervised staging or recover REM reliably on strict holdout (REM recall $\sim 3\%$). Biological interpretation of higher- K population sweeps will further test whether MSSV supports the same profound substage structure found in single-laboratory cGMVAE work [2].

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A Data and cohort

A.1 MSSV cohort and quality screening

MSSV (OpenNeuro ds006366) [4] comprises invasive EEG and neck EMG from 92 mice across five European laboratories, with expert Wake, NREM, REM, and Artifact labels on a 4 s grid for evaluation only. This report analyses laboratories 2 (Bern), 3 (Copenhagen), and 5 (Lyon): healthy wild-type sites with complementary montages, excluding laboratories 1 and 4 (pathology-specific or single-electrode cohorts).

Quality screening retains mice with low EEG drift and EMG that separates Wake from NREM. Drift is quantified as relative 0.5–2 Hz power ≤ 0.2 of 0.5–30 Hz on each retained electrode. Wake/NREM separability uses a root-mean-square ratio ≥ 2 on neck EMG. The screened cohort comprises 20 mice (6 Bern, 10 Copenhagen, 4 Lyon) with 42 recording runs and ~ 453 h of EEG/EMG ($\sim 4.1 \times 10^5$ artefact-free 4 s epochs after screening). Training mice never appear in the validation fold for a given split (S9 Table). No new animal procedures were performed.

S8 Table. Quality-screened cohort inventory. Per-mouse recording runs, hours, and 4 s epoch counts (MSSV annotation grid).

Mouse	Site	Runs	Hours	Epochs (4 s)
sub-071	Bern (2)	2	12	10,800
sub-072	Bern (2)	2	12	10,800
sub-076	Bern (2)	2	12	10,800
sub-077	Bern (2)	2	13	11,700
sub-080	Bern (2)	2	12	10,800
sub-081	Bern (2)	2	12	10,800
sub-038	Copenhagen (3)	2	48	43,201
sub-039	Copenhagen (3)	3	12	10,836
sub-041	Copenhagen (3)	3	12	10,803
sub-043	Copenhagen (3)	1	24	21,600
sub-048	Copenhagen (3)	3	12	10,804
sub-054	Copenhagen (3)	2	48	43,194
sub-056	Copenhagen (3)	1	23.6	21,196
sub-059	Copenhagen (3)	1	24	21,600
sub-060	Copenhagen (3)	1	8	7,212
sub-069	Copenhagen (3)	1	24	21,600
sub-087	Lyon (5)	6	72	64,788
sub-088	Lyon (5)	2	24	21,595
sub-089	Lyon (5)	2	24	21,596
sub-092	Lyon (5)	2	24	21,595
Total (20 mice)		42	452.6	407,320

A.2 Preprocessing and spectral inputs

Montages are harmonised to a parietal-frontal EEG differential plus neck EMG (laboratory 2: EEG1 and EEG4; laboratories 3 and 5: EEG1 and EEG2). Recordings are segmented into 4 s epochs at 128 Hz. Artifact epochs are removed. Each epoch is a log-power spectral map $\mathbf{x}_t \in \mathbb{R}^{C \times F}$, batched into contiguous sequences for temporal models without hand-crafted expert features.

Per-laboratory channel sets were fixed from incohort ablation winners (full sweep outcomes in Incohort performance ceiling). Spectral passbands are summarised in Spectral bands and frequency content.

A.3 Spectral bands and frequency content

Each 4 s epoch comprises 512 samples at 128 Hz sampling rate. The CNN input is a log-power spectrum from an unwindowed real FFT (frequency resolution 0.25 Hz; Nyquist 64 Hz). Band limiting is applied in the frequency domain on the complex spectrum before log-power: a null lower cutoff retains all bins from DC upward, and an upper cutoff zeroes bins above the configured limit. The model therefore receives the full spectral shape inside each passband, not a small vector of hand-defined sleep bands.

Locked per-laboratory passbands (holdout protocol).

Site	EEG channels	EEG passband	EMG passband
Bern (lab 2)	EEG1, EEG4	0–30 Hz	3–100 Hz ($\leq$ 64 Hz Nyquist)
Copenhagen (lab 3)	EEG1, EEG2	0–20 Hz	5–60 Hz
Lyon (lab 5)	EEG1, EEG2	0–20 Hz	5–60 Hz

Bern uses no per-epoch post-normalisation on log-power spectra. Copenhagen and Lyon apply per-epoch z-scoring of log-power after band limiting (post-normalisation). These limits were chosen from incohort ablations (Incohort performance ceiling) and held fixed for all cross-laboratory holdout runs.

Diagnostic and physiology bands. Quality screening compares relative 0.5–2 Hz EEG power to 0.5–30 Hz total (MSSV cohort and quality screening). For substage physiology panels (Fig 3, S1 Fig) and summary tables (S3 Table), log-power spectra are aggregated into interpretable bands on the same frequency grid: delta 0.5–4 Hz, theta 4–8 Hz, and broadband EMG power summed over the retained EMG passband. Reported θ/δ ratios use these bands.

A.4 Cross-validation protocol

Folds were designed so each site contributes holdout mice every fold while balancing validation load: fold 1 is epoch-heaviest ($\sim 32\%$ of cohort epochs in validation, driven by long Copenhagen and Lyon recordings) whereas folds 2–4 hold out 21–25% of epochs each. sub-069 is assigned to fold 2 rather than fold 4 so Copenhagen does not contribute four simultaneous holdout mice on the substage-analysis fold. Each VAE configuration trains three random seeds per fold. Reported metrics use the best prior NMI among seeds.

S9 Table. Four-fold leave-mice-out assignments. Top: held-out mice per site and fold. Bottom: validation versus training pool sizes (joint scope trains on all non-held-out mice across sites).

Fold	Site	Held-out mice	Runs	Hours	Epochs
1	Bern (2)	sub-071	2	12	10,800
	Copenhagen (3)	sub-038, sub-039	5	60	54,037
	Lyon (5)	sub-087	6	72	64,788
2	Bern (2)	sub-072, sub-076	4	24	21,600
	Copenhagen (3)	sub-041, sub-048, sub-069	7	48	43,207
	Lyon (5)	sub-088	2	24	21,595
3	Bern (2)	sub-077	2	13	11,700
	Copenhagen (3)	sub-043, sub-054	3	72	64,794
	Lyon (5)	sub-089	2	24	21,596
4	Bern (2)	sub-080, sub-081	4	24	21,600
	Copenhagen (3)	sub-056, sub-059, sub-060	3	55.6	50,008
	Lyon (5)	sub-092	2	24	21,595

Fold	Val mice	Val runs	Val hours	Val epochs	Train mice	Train epochs	Val (%)
1	4	13	144	129,625	16	277,695	31.8
2	6	13	96	86,402	14	320,918	21.2
3	4	7	109	98,090	16	309,230	24.1
4	6	9	103.5	93,203	14	314,117	22.9

B Training, checkpoints, and metrics

B.1 Locked hyperparameters

Latent dimension $d=6$ (matched to thesis-scale mixture models), embedding dimension 8, learning rate 1.3×10^{-3} , batch size 128, 200 epochs, three random seeds per fold. Sequence length $T=64$ epochs (~ 4.3 min); sticky self-transition weight $\kappa=0.92$. CNN channel widths [32, 64, 128, 128], kernel sizes [7, 5, 5, 3], strides 2; MLP widths [512, 256, 128] (encoder) and [128, 256, 512] (decoder). Full YAML paths: Supplementary experimental settings.

B.2 Incohort ablation rationale

Per-laboratory spectral bandpass, montage, post-normalisation, and MLP width were fixed from incohort sweeps before any holdout run (Preprocessing and spectral inputs, Incohort performance ceiling). These runs pool all mice within a site and estimate a performance *ceiling*; they do not use leave-mice-out folds.

B.3 Sensitivity checks

On joint holdout fold 4, shortening the sequence to $T=32$ without the warm sticky schedule collapsed prior NMI (best seed ~ 0.17), whereas $T=64$ with warm training reached ~ 0.65 . On laboratory 3 incohort, warm-started HMM-GMM beat a cold HMM-GMM start (best-of-three prior NMI 0.734 vs. 0.694); gains were smaller on laboratories 2 and 5, where simpler priors remained competitive. We did not grid-search every knob at holdout; locked settings prioritise reproducibility.

B.4 Warm prior schedule (cHMM-GMVAE)

Epochs 1–17: standard Gaussian KL on the encoder. Epoch 18: K -means on encoder means initialises GMM emissions. Epochs 18–36: independent mixture KL. From epoch 37: sticky HMM-GMM transitions ramp in over 18 epochs [16, 17]. β is 0 for the first ten epochs, then ramps linearly from 0.01 to 1.0.

B.5 Staging on encoder means

At holdout, emissions use μ_t rather than sampled $\mathbf{z}_t$. The posterior mean can bias emission scores slightly relative to stochastic latents but reduces Monte Carlo variance in Viterbi decoding and improves reproducibility across seeds once the encoder is trained.

B.6 Unsupervised checkpoint score

Expert labels are not used to select weights during training. Let $\tilde{\omega}(\varepsilon)$ denote validation log-likelihood (sequence-normalised negative ELBO) and $H_{\text{norm}}(\hat{Y})$ the entropy of the predicted state histogram normalised by $\log K$ ($0 = \text{collapse to one state}$, $1 = \text{uniform occupancy}$). Following the thesis composite criterion [2],

$$S(\varepsilon) = \beta_{\text{ckpt}} \tilde{\omega}(\varepsilon) + (1 - \beta_{\text{ckpt}}) H_{\text{norm}}(\hat{Y}), \quad \beta_{\text{ckpt}} = 0.9. \quad (5)$$

The epoch maximising $S(\varepsilon)$ after a 5% warmup is retained per seed. Prior NMI against expert macros is computed only *after* checkpoint selection for reporting.

B.7 Macro-aligned accuracy and label switching

Macro accuracy maps each predicted cluster to its dominant expert macro label (Hungarian alignment when $K=3$), then reports the fraction of epochs matching Wake/NREM/REM. Accuracy can exceed chance and diverge from NMI when $K > 3$ substates collapse onto the same macro label, when too few clusters are active, or when clusters are macro-pure but permuted. Mixture components are identifiable only up to permutation; NMI is invariant to permutations, but cluster indices differ across seeds unless post-hoc alignment is applied for tables and figures.

B.8 Seed aggregation and inferential tests

Each fold trains three seeds; holdout prior NMI is evaluated at each seed’s $S(\varepsilon)$ checkpoint; the maximum across seeds is the headline fold score (Tables 1–2, S11 Table). Paired fold-level $\Delta\text{NMI} = \text{NMI}_{\text{cHMM}} - \text{NMI}_{\text{cGMVAE}}$ uses the same held-out mice ($n=4$ folds). We report mean Δ , bootstrap 95% CI (10 000 resamples), one-sided Wilcoxon (H_0 : median $\Delta \leq 0$), and paired t -test as sensitivity (S12 Table, S8 Fig). Fold 4 per-mouse paired tests are a secondary unit-of-analysis check (S2 Fig).

B.9 Prior log-likelihood at holdout

Holdout prior log-likelihood complements NMI when judging unsupervised fit (S14 Table). For **cGMVAE** ($T=1$), each 4s epoch is treated independently: the reported value is the mean $\log p(z_t)$ under the static GMM prior on encoder means (metrics.txt Likelihood). For **HMMGMVAE** and **cHMM–GMVAE** ($T=64$), windows of 64 contiguous epochs are scored with the HMM forward marginal: metrics.txt reports mean $\log p(z_{1:T})$ per sequence ($\log p(\mathbf{z}_{1:T})$). Because the sequence length differs, raw totals are *not* comparable across cGMVAE and sequence-prior models. The normalised column divides cGMVAE values by $L=6$ (per-latent-dimension mean log-density at $T=1$) and sequence models by $T \cdot L$ (matching the K-sweep axis in Fig 2). Higher normalised values indicate better prior density on held-out latents; they need not track NMI when expert macros misalign with mixture components.

C Prior models and protocol ablations

C.1 HMM–GMM prior (equations)

Each discrete latent state $s_t \in \{1, \dots, K\}$ emits a diagonal Gaussian in $\mathbb{R}^d$. The per-step emission log-density is

$$\log b_k(\mathbf{z}_t) = \log \mathcal{N}(\mathbf{z}_t \mid \mathbf{m}_k, \text{diag}(\mathbf{v}_k)). \quad (6)$$

Sticky transitions use self-transition weight $\kappa=0.92$. The marginal log-likelihood of a latent window is

$$\log p(\mathbf{z}_{1:T}) = \log \sum_{s_{1:T}} \pi_{s_1} \prod_{t=2}^T A_{s_{t-1}, s_t} \prod_{t=1}^T b_{s_t}(\mathbf{z}_t), \quad (7)$$

and the sequence KL term averaged over time is

$$\mathcal{L}_{\text{KL}} = \frac{1}{T} \sum_{t=1}^T [\log q_\phi(\mathbf{z}_t \mid \mathbf{x}_t) - \log p(\mathbf{z}_{1:T})]. \quad (8)$$

Training schedule, warm-start phases, and CNN widths are in Training, checkpoints, and metrics.

C.2 Raw time-domain HMM failure

Raw time-domain HMM failure. The MSc thesis showed that classical HMMs on raw MSSV EEG/EMG voltages do not generalise across subjects: population prior NMI collapses while expert-feature inputs remain usable (thesis Ch. 5). On the locked cross-lab protocol, joint HMM (raw) prior NMI is near chance (mean 0.054; fold 4 peak 0.21), motivating the spectral CNN front-end used throughout this report. End-to-end CNN encoders on filtered raw traces are discussed in Future directions.

C.3 Decoder-only vs encoder+decoder conditioning

The MSc thesis default fed subject embeddings to *both* encoder and decoder on laboratory 3 only [2]. Before holdout, we compared decoder-only and encoder+decoder paths on locked incohort recipes at all three sites ($K=3$, three seeds):

Site	Decoder-only	Encoder+decoder	Best-of-three prior NMI
Bern (lab 2)	0.593	0.578	Decoder-only
Copenhagen (lab 3)	0.737	0.621	Decoder-only
Lyon (lab 5)	0.534	0.430	Decoder-only

All three sites favoured decoder-only conditioning on incohort prior NMI. The thesis headline (~ 0.70 population NMI) used encoder+decoder on laboratory 3 with expert features and leave-one-subject-out evaluation [2], which is not directly comparable to these three-lab spectral incohort runs. Holdout fold-level encoder+decoder comparisons were not run under the locked paper protocol; the main line uses decoder-only throughout.

C.4 Incohort performance ceiling

Before cross-laboratory holdout, we ran extensive within-laboratory sweeps over preprocessing, CNN-VAE architecture, prior tier (static GMM vs. HMM-GMM vs. warm sticky HMM-GMM), and sequence length T . All mice in a laboratory were pooled for training and validation tuning (no leave-mice-out). Reported values are best-of-three prior NMI at $K=3$ with decoder-only conditioning. These runs fixed the per-site recipes in Preprocessing and spectral inputs and estimate an upper performance bound when site-specific tuning is allowed. They are *not* the locked holdout protocol in Table 2. Macro accuracy was not logged systematically in the incohort sweeps.

Bern (laboratory 2): cGMVAE reached 0.593 with EEG1+EEG4, no post-normalisation, and EEG 0–30 Hz. Temporal priors did not beat the static mixture at $T=1$ (best cHMM-GMVAE 0.571 with warm prior and 50 Hz notch). Copenhagen (laboratory 3): cGMVAE peaked at 0.737; warm cHMM-GMVAE at $T=64$ matched within 0.003 (0.734). Lyon (laboratory 5): cHMM-GMVAE with EMG-wide notch preprocessing at $T=32$ reached 0.640 versus 0.534 for cGMVAE, though only one of three seeds was healthy at that peak.

Incohort ceiling prior NMI. Best-of-three prior NMI and winning configuration per laboratory and model.

Lab	Model	Best NMI	Winning lock (summary)
2	cGMVAE	0.593	EEG1+EEG4, no postnorm, EEG 0–30 Hz, EMG 3–100 Hz
2	cHMM-GMVAE	0.571	warm prior + 50 Hz notch, below cGMVAE at $T=1$
3	cGMVAE	0.737	postnorm, EEG 0–20 Hz, wide MLP
3	cHMM-GMVAE	0.734	warm HMM-GMM, $T=64$, same prepro/arch
5	cGMVAE	0.534	postnorm, EEG 0–20 Hz, wide MLP
5	cHMM-GMVAE	0.640	EMG wide + notch, $T=32$ (1/3 seeds healthy)

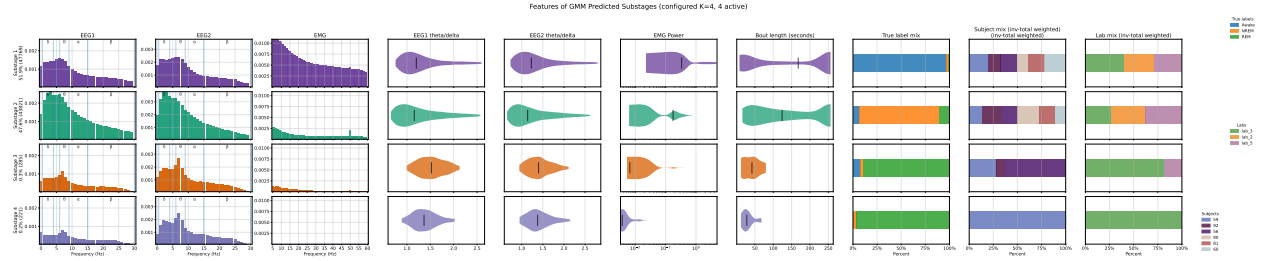
Ceiling reference only. Holdout within-lab NMI and accuracy under the same locked recipes appear in S2 Table. Joint holdout remains below these ceilings because leave-mice-out validation is strictly harder than pooling all laboratory mice during tuning.

C.5 Synthetic HMM validation

Controlled synthetic datasets confirmed that MAR-HMM recovers temporal structure when discriminative information is dynamic rather than marginal (thesis Ch. 3, Table 7 therein). That step validated optimisation and inference before MSSV experiments; it is not repeated here.

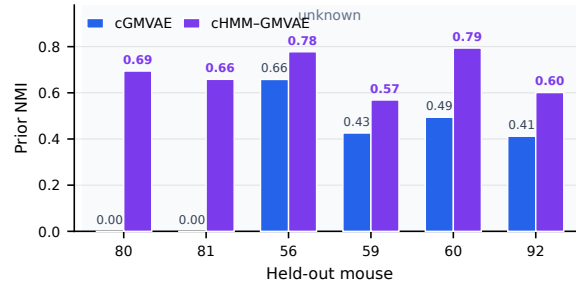
D Supplementary figures

D.1 S1 Fig.



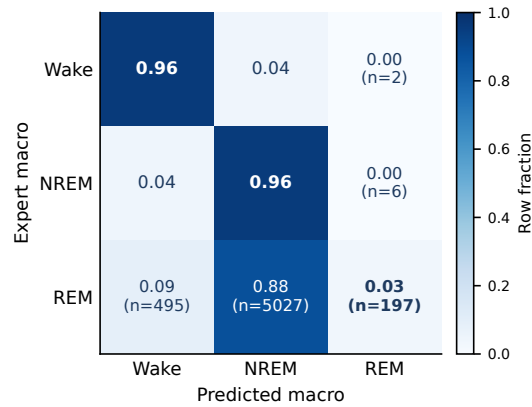
Substage physiology ($K=4$, fold 4 holdout). EEG/EMG power spectra, θ/δ and bout-length distributions, and expert macro composition per discovered state.

D.2 S2 Fig.



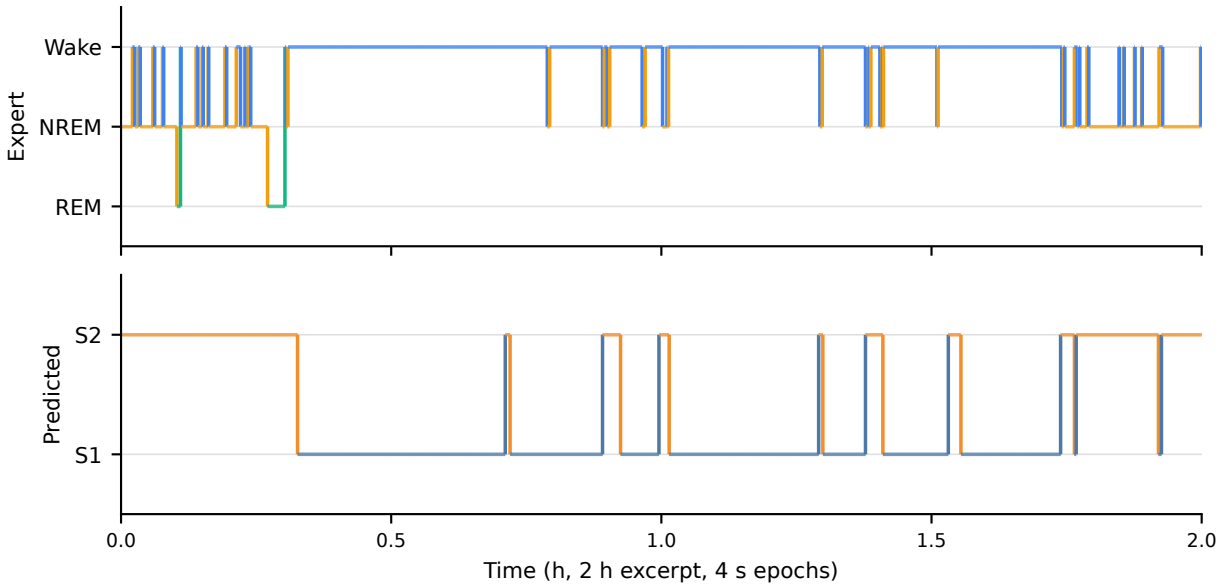
Per-mouse prior NMI on fold 4. Paired bars for cGMVAE and cHMM-GMVAE on each held-out mouse.

D.3 S3 Fig.



Macro confusion (fold 4, $K=3$, cHMM-GMVAE, best seed). Row-normalised 3×3 matrix: expert Wake/NREM/REM (rows) vs. predicted macro after Hungarian alignment (columns). REM row shows low recall ($\sim 3\%$) on this holdout fold.

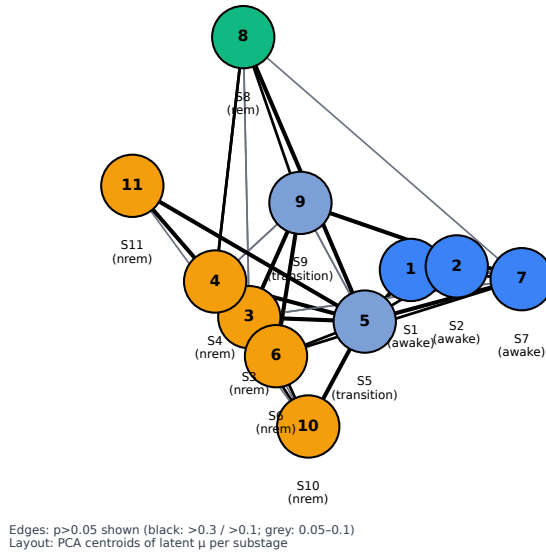
D.4 S5 Fig.



Hypnogram exemplar (fold 4 holdout, $K=4$, best seed). Thesis-style dual track: expert Wake/NREM/REM annotation (top) versus predicted substages (bottom) on a 2 h excerpt.

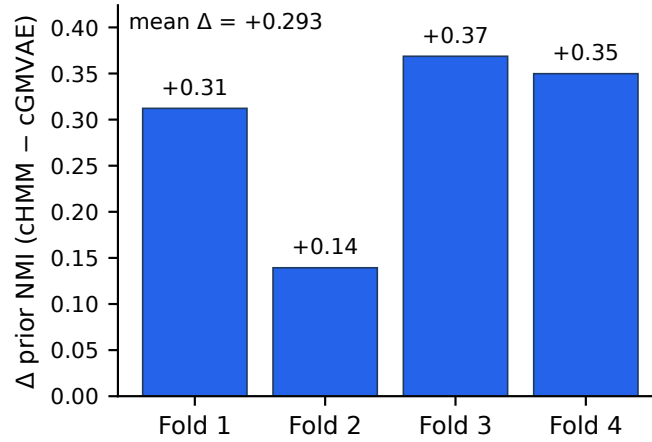
D.5 S6 Fig.

Fold 4 $K=13$ cHMM-GMVAE taxonomy (thesis-like)



Finer-resolution taxonomy ($K=13$, fold 4 holdout). Illustrative transition map at thesis-scale resolution. Holdout NMI 0.525 vs. 0.643 at $K=4$.

D.6 S8 Fig.



Paired fold-level gain (best-of-three prior NMI). $\Delta\text{NMI} = \text{NMI}_{\text{cHMM-GMVAE}} - \text{NMI}_{\text{cGMVAE}}$ on the same held-out mice per fold.

E Supplementary tables

E.1 S2 Table.

Per-fold holdout by site (Table 2 detail). Each cell is prior NMI / macro accuracy (best-of-three seeds). Joint block: population training, metrics on held-out mice from the listed site only. Within block: train and evaluate inside the same site. Abbreviations: Raw = HMM (raw), cGM = cGMVAE, HGM = HMMGMVAE, cHMM = cHMM-GMVAE.

Joint training (evaluate on held-out site mice only)

Fold	Site	Raw	cGM	HGM	cHMM
1	Bern	—	0.00/0.52	0.00/0.52	0.67/0.89
	Cph	—	0.40/0.74	0.64/0.87	0.71/0.92
	Lyon	—	0.20/0.66	0.44/0.84	0.52/0.87
2	Bern	—	0.25/0.69	0.16/0.67	0.26/0.77
	Cph	—	0.35/0.78	0.34/0.73	0.74/0.93
	Lyon	—	0.20/0.66	0.29/0.73	0.48/0.90
3	Bern	—	0.64/0.90	0.00/0.50	0.65/0.90
	Cph	—	0.41/0.79	0.68/0.88	0.71/0.93
	Lyon	—	0.29/0.74	0.61/0.86	0.62/0.88
4	Bern	0.18/0.66	0.00/0.49	0.00/0.49	0.69/0.91
	Cph	0.24/0.73	0.51/0.87	0.69/0.92	0.66/0.91
	Lyon	0.24/0.72	0.43/0.81	0.59/0.85	0.59/0.86

Within-site training

Fold	Site	Raw	cGM	HGM	cHMM
1	Bern	0.00/0.52	0.66/0.89	0.70/0.90	0.70/0.90
	Cph	0.05/0.58	0.55/0.88	0.58/0.89	0.76/0.94
	Lyon	0.00/0.50	0.47/0.86	0.58/0.89	0.54/0.88
2	Bern	0.00/0.47	0.27/0.74	0.30/0.76	0.34/0.75
	Cph	0.00/0.47	0.31/0.73	0.62/0.88	0.75/0.94
	Lyon	0.00/0.54	0.39/0.83	0.53/0.88	0.50/0.90
3	Bern	0.00/0.50	0.63/0.89	0.66/0.90	0.67/0.90
	Cph	0.03/0.58	0.60/0.87	0.69/0.89	0.73/0.93
	Lyon	0.00/0.48	0.50/0.86	0.61/0.86	0.60/0.88
4	Bern	0.02/0.52	0.64/0.89	0.69/0.90	0.67/0.90
	Cph	0.04/0.63	0.60/0.90	0.66/0.93	0.72/0.94
	Lyon	0.00/0.46	0.53/0.86	0.62/0.90	0.62/0.88

E.2 S3 Table.

Substage summary ($K=4$, fold 4 holdout). Physiology labels, dominant expert macro composition, θ/δ , bout length, and occupancy.

State	Physiology label	Dominant macro (%)	θ/δ	Median bout (s)	Occupancy (%)
S1	Wake-dominant	97	1.47	16	51.9
S2	NREM-dominant	83	1.22	114	47.6
S3	REM-sparse A	89	1.48	42	0.3
S4	REM-sparse B	96	1.47	28	0.2

Occupancy-ordered states ($K=4$, seed 3). Rare REM-dominated clusters (S3, S4) carry $< 1\%$ occupancy. Transition-rich structure at higher K is in S6 Fig.

E.3 S4 Table.

Joint holdout macro accuracy. Best-of-three-seed NMI and macro-aligned accuracy by fold for cGMVAE and cHMM-GMVAE ($K=3$). Clusters mapped to dominant expert macro label when $K > 3$.

Fold	cGMVAE NMI	cGMVAE Acc.	cHMM NMI	cHMM Acc.
1	0.267	0.658	0.579	0.885
2	0.256	0.724	0.396	0.818
3	0.307	0.744	0.675	0.913
4	0.299	0.766	0.649	0.900
Mean	0.282	0.723	0.575	0.879

E.4 S5 Table.

K-sweep macro accuracy (fold 4). Best-of-three-seed NMI and macro-aligned accuracy versus substage count K . Same mapping rule as S4 Table.

K	NMI	Macro accuracy
3	0.626	0.900
4	0.402	0.796
5	0.523	0.925
6	0.562	0.910
7	0.577	0.899
8	0.509	0.891
9	0.437	0.880
10	0.444	0.901
11	0.499	0.890
12	0.482	0.872
13	0.489	0.883
14	0.461	0.882
15	0.438	0.881

E.5 S1 Table.

Literature comparison. Supervised ceilings, transductive unsupervised, and static unsupervised methods versus this work. Complementary MSSV work includes QSLP-AE cross-species alignment [18] and manual-scoring consensus benchmarks [5].

Method	Setting	Metric	Notes
SPINDLE	Multi-lab supervised	High accuracy	End-to-end CNN; cross-lab extrapolation with supervised retraining [6]
REST	Supervised Transformer	$\kappa \approx 0.87$	Automated mouse staging on multi-strain cohorts [7]
García Ciudad et al.	Supervised temporal context	Accuracy	~28s context improves staging vs. single epoch; REM gains largest [13]
AccuSleep / SegWay	Transductive unsupervised	Accuracy	Per-mouse calibration baseline [8, 9]
mcRBM	Pooled unsupervised	Substages	No cross-lab holdout protocol [1]
This work	Joint zero-shot holdout	Prior NMI 0.58	Decoder-only cHMM-GMVAE, $K=3$ macro ladder

E.6 S10 Table.

Per-class classification metrics (fold 4, $K=3$, cHMM-GMVAE, best seed). Cohen’s κ , macro F1, and per-class recall after Hungarian alignment.

Run	κ	Macro F1	Wake rec.	NREM rec.	REM rec.
fold4 cHMM K=3	0.811	0.638	0.959	0.955	0.034

E.7 S11 Table.

Seed stability (joint holdout). Prior NMI mean, standard deviation, and best-of-three per fold and model.

Fold	Model	Seeds	NMI mean	NMI SD	NMI best
1	cGMVAE	2	0.219	0.048	0.267
1	cHMMGMVAE	3	0.521	0.041	0.579
2	cGMVAE	2	0.168	0.089	0.256
2	cHMMGMVAE	3	0.375	0.014	0.396
3	cGMVAE	2	0.297	0.009	0.307
3	cHMMGMVAE	3	0.450	0.315	0.675
4	cGMVAE	2	0.214	0.085	0.299
4	cHMMGMVAE	3	0.581	0.056	0.649

E.8 S12 Table.

Paired inferential tests (cHMM-GMVAE vs. cGMVAE, joint holdout). Primary: best-of-three NMI per fold; sensitivity: seed-mean NMI per fold; fold 4: per-mouse paired test. Wilcoxon: one-sided H_0 : median $\Delta \leq 0$; bootstrap 95% CI on mean Δ over resampled folds ($B=10\,000$).

Comparison	Unit	Δ mean	95% CI	p (Wilcoxon)
Primary	4 folds	0.293	[0.192, 0.359]	0.062
Sensitivity	4 folds (seed mean)	0.258	[0.180, 0.335]	0.062
Fold 4 mice	6 mice	0.351	[0.173, 0.538]	0.016

E.9 S14 Table.

Holdout prior log-likelihood (joint scope, $K=6$, best-of-three seed). **cGMVAE** uses sequence length $T=1$: each 4s epoch is scored independently with the static GMM prior (mean $\log p(z_t)$ per epoch; Likelihood in run metrics). **HMMGMVAE** and **cHMM-GMVAE** use $T=64$ contiguous epochs per window; the reported value is the HMM forward marginal mean $\log p(z_{1:T})$ per sequence ($\log p(z_{1:T})$). Raw totals are *not* comparable across $T=1$ and $T=64$ models because sequence models accumulate density over 64 steps. The normalised column divides cGMVAE by $L=6$ and sequence models by $T \cdot L=384$, matching the orange axis in Fig 2 for $T=64$ runs. Higher values indicate better prior density on held-out encoder means; they need not track NMI when mixture structure misaligns with expert macros. Fold 4 cGMVAE (seed 2) shows an extreme negative raw value despite moderate NMI—a known case where prior density and clustering quality diverge.

Fold	Model	T	Holdout NMI	Prior LL (raw)	Norm.
1	cGMVAE	1	0.267	-17.59	-2.931
1	HMMGMVAE	64	0.409	-3755.5	-9.780
1	cHMM-GMVAE	64	0.579	-26.8	-0.070
2	cGMVAE	1	0.256	-11.86	-1.976
2	HMMGMVAE	64	0.236	-1200.4	-3.126
2	cHMM-GMVAE	64	0.396	-307.3	-0.800
3	cGMVAE	1	0.307	-12.51	-2.085
3	HMMGMVAE	64	0.497	-166.0	-0.432
3	cHMM-GMVAE	64	0.675	-84.5	-0.220
4	cGMVAE	1	0.299	-68904.82	-11484.137
4	HMMGMVAE	64	0.404	-264.6	-0.689
4	cHMM-GMVAE	64	0.649	281.1	0.732

F Thesis context and future directions

F.1 Relation to the MSc thesis

The accompanying MSc thesis [2] developed unsupervised mouse sleep staging with mixture priors on variational autoencoder latents. That work used a *single* MSSV laboratory (Copenhagen, laboratory 3): a curated ten-mouse subset, expert-engineered spectral features for classical HMM baselines, and **encoder+decoder** subject conditioning (embeddings in both encoder and decoder). Population leave-one-subject-out prior NMI reached about 0.70 when staging from the cGMVAE mixture prior; substage discovery at higher mixture counts used the same in-laboratory protocol (thesis Figs 27–30).

This report changes three elements for cross-laboratory evaluation. First, **decoder-only** conditioning: subject embeddings enter the decoder only so held-out mice can be staged from encoder means and the mixture prior without subject identifiers (Decoder-only vs encoder+decoder conditioning). Second, a **quality-screened 20-mouse cohort** from laboratories 2, 3, and 5 with per-site spectral preprocessing locks (Preprocessing and spectral inputs) and four-fold leave-mice-out cross-validation (Tables 1–2). Third, a **spectral CNN front-end** and end-to-end warm sticky HMM-GMM prior replace the thesis two-step pipeline (cGMVAE latents followed by MAR-HMM decoding on expert features).

Synthetic HMM validation and raw time-domain failure modes from the thesis are summarised in Synthetic HMM validation and Raw time-domain HMM failure. Planned extensions (laboratory conditioning, richer subject/lab disentanglement, and raw-signal CNN encoders) are outlined in Future directions.

F.2 Future directions

Laboratory and subject conditioning. This report uses *subject* embeddings in the decoder only (Decoder-only vs encoder+decoder conditioning). Preliminary cv4fold experiments with *subject_lab* embeddings (concatenating laboratory identity alongside mouse identity) collapsed on joint holdout (prior NMI near zero), suggesting that explicit lab tags did not stabilise cross-site transfer under the locked recipe. A middle ground worth testing is hierarchical or factorised conditioning: separate embeddings for acquisition site and animal, or adversarial removal of lab information from encoder latents while retaining subject-specific reconstruction in the decoder. Such designs may disentangle montage and scoring context from individual variability more cleanly than a single subject index when Bern, Copenhagen, and Lyon contribute unequal cohort sizes (S8 Table).

End-to-end CNN on raw EEG/EMG. The locked pipeline maps each 4 s epoch to a log-power spectral tensor via FFT and trains a CNN-VAE on that representation. The thesis showed that classical HMMs on *raw* time-domain MSSV inputs fail to generalise across subjects (Raw time-domain HMM failure); this report’s HMM (raw) holdout baseline remains near chance on the cross-lab protocol (joint mean prior NMI 0.054). An open direction is to replace the hand-crafted spectral front-end with a convolutional encoder on band-pass filtered raw EEG/EMG traces, closer to supervised mouse CNN classifiers [6, 13], while retaining the same decoder-only conditioning and sticky mixture prior at the latent level. Fair comparison would require matched context length (T epochs), preprocessing locks per laboratory, and the same leave-mice-out folds; spectral and raw encoders need not share identical capacity.

Substage selection and biology review. Mixture count K was chosen on holdout fold 4 only (Limitations). Train-only or nested cross-validation for K , plus expert review of substage names from physiology panels (as in mcRBM [1]), would strengthen claims beyond macro NMI.

F.3 Supplementary experimental settings

Full hyperparameters for joint and within-laboratory holdout models are provided as supplementary material.